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r/Python • 8 mo. ago
commandlineuser

pandas 2.0 and the Arrow revolution

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code_mc • 8 mo. ago

It's quite amazing to see the synergy between the pandas and polars creators. I really didn't expect to see the presented example tbh.

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jorge1209 • 8 mo. ago

The original author of pandas is the co-creator of arrow.

Arrow is Wes McKinney's attempt to fix some back end issues with Pandas, but Pandas still has to deal with the mistakes made in the front-end API design. Polars gets to leverage McKinney's improvements to the back-end while providing a cleaner front-end.

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midnitte • 8 mo. ago

Which really makes this point rather funny (I took it to just be in jest):

> Besides just ignore Polars and use pandas

10 Share

jorge1209 • 8 mo. ago

It was clearly very much in jest. The entire objective of arrow is to enable this kind of data interchange. You aren't tied down to any one particular analytics engine, but can pick the best tool for the job.

There are some things that polars will be much better at than pandas, and there are some things pandas will continue to do better than polars.

With arrow you can pick the best tool for the job, but don't have to worry that in doing so you introduce time consuming and expensive steps that do nothing but copy memory around from one engines format to the others.

2 Share

CrackerJackKittyCat • 8 mo. ago

Yeah, Polars smells like C# to Pandas's Java. In a good way!

24 Share

tinkr_ • 8 mo. ago

Yeah, it's pretty rare to see cooperation between two projects that occupy a similar product space. Usually it's when both projects are run more for passion than some type external reward.

Another place I've seen this recently is with Neovim. During NeovimConf last year they literally invited the creators of multiple other competing modal editors like Helix to give presentations on what their editors offer and why Neovim users should try them.

In the free software community we're all friends. :) Our mission is to provide tools that are available to anyone. As a pandas core developer I'm happy to also contribute to Polars, and I'm happy to see it succeed. It solves things that pandas can't address, and for many use cases it's an improvement. For many others, pandas is still a better option. Polars is not as well tested as pandas, and it's mostly a one-person project.

I hope in the future we can share more code with Polars. It would be good to have I/O connectors, or the plotting extensions now in pandas being independent, and working for both projects, and other such as Dask, Vaex, Koalas...

So, different project, but same team. :)

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gopietz • 8 mo. ago

I'm currently porting a data processing application from pandas to polars and while there's still a few things missing, it has been a really enjoyable process.

Don't get me wrong, pandas is great, but i'm starting to believe that we have reached a point where a complete rewrite like polars might actually work out better than teaching an old dog new tricks.

It feels a bit like when tensorflow 2.0 tried to make everything feel more like pytorch. Most people were much happier just using pytorch instead and leaving the old baggage behind.

In my experience, polars as a drop-in replacement is 7-10x faster. If you really optimize your pipeline to the polars mindset, it improves to 50-100x. It's stupidly fast.

One of two things will happen to pandas: a) they will never reach this form of acceleration. b) they use polars as a backend and rebuild functionality that is currently missing.

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ertgbnm • 8 mo. ago

There is a deep irony to the fact that I spent more time reading this post than pyarrow will probably ever save me given how small the stuff I work on is. Still really cool to see what's going on out there though.

30 Share ...

whiterabbitobj • 8 mo. ago

But that's the programming way of life... spend 8 hours automating a 5s daily task.

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ecapoferri • 8 mo. ago

OP, thanks so much for posting this. This seems like crucial reading for anyone who does anything with data in Python. I'm on subs like this expressly to learn and stay on top of New developments.

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magnetichira • 8 mo. ago

Very interesting.

The benefits seem mostly contained to string and datetime formats.

Does the arrow backend have any limitations with regards to floating point data, like for example interacting it with numpy/scipy?

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WoodenNichols • 8 mo. ago

Not certain I understand. Someone created a Python library called arrow? One that clears up/minimizes issue with pandas.

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blewrb • 8 mo. ago

Arrow is a library for a format for storing columnar data in memory and functions for operating on said data, written in C. It can be used from various languages, including Python.

Arrow was written primarily by Wes McKinney, original author of Pandas, as a result of the pain points he encountered with in-memory data storage while writing Pandas. Polars was designed to use Arrow for its data, and Pandas 2 can now also optionally use Arrow as its in-memory data storage backend.

Wes's vision is/was that Arrow would become the lingua franca for columnar data, making accessing and operating on the same data trivial between e.g. R and Python. It's even used on GPUs for GPU-based data frame libraries..

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heartofcoal • 8 mo. ago

Arrow is an Apache product

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Yoshimi917 • 8 mo. ago

pandas ily

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